

MIRNY STATION, Antarctica

WMO: 895920

Lat: 66.553S

Lon: 93.010E

Elev: 147

StdP: 14.62

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-21.7	-18.3	-29.5	1.1	-20.6	-26.5	1.3	-17.4	98.1	9.0	85.4	-0.4	18.1	160	1.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	35.3	31.9	34.1	30.9	32.9	30.1	32.4	34.4	31.5	33.4	30.6	32.5	9.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.6	24.9	32.4	29.5	23.7	31.9	28.3	22.4	31.2	12.1	34.8	11.7	33.6	11.3	32.6	36.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 84.4	70.6	57.5	-25.9	38.2	4.9	2.6	-29.4	40.1	-32.3	41.7	-35.1	43.2	-38.7	45.1		
(5)			-26.2	33.8	4.8	1.5	-29.6	34.9	-32.4	35.8	-35.1	36.7	-38.6	37.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	27.9	21.7	12.6	7.1	3.9	3.9	2.5	1.8	2.9	7.4	19.3	26.8		
	DBStd	12.42	3.69	5.28	7.20	8.66	9.75	9.78	10.66	9.74	9.88	7.57	6.09	3.78		
	HDD50	14073	685	792	1158	1288	1429	1382	1472	1495	1413	1320	920	719		
	HDD65	19548	1150	1212	1623	1738	1894	1832	1937	1960	1863	1785	1370	1184		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	19.0	12.5	16.1	17.1	22.9	22.3	25.3	23.3	20.4	18.9	18.5	17.5	13.8		
	PrecAvg	21.1	0.8	0.9	1.4	1.5	2.2	2.6	2.6	2.5	2.6	1.9	1.6	0.9		
	PrecMax	39.7	3.9	2.5	3.4	4.6	7.6	7.4	10.1	7.5	9.4	6.3	7.8	3.6		
(15) Precipitation	PrecMin	6.3	0.0	0.0	0.0	0.0	0.1	0.2	0.5	0.1	0.1	0.0	0.0	0.0		
	PrecStd	9.4	0.8	0.7	1.0	1.2	1.7	2.0	2.2	1.7	2.0	1.5	1.6	0.9		
	DB	37.2	35.1	29.3	27.9	27.2	24.9	26.6	26.0	25.3	25.1	33.0	37.0			
	MCWB	33.1	31.6	27.9	27.5	26.3	23.5	25.7	25.3	24.3	23.9	29.0	32.9			
	2% DB	35.6	33.2	26.9	24.3	23.0	22.5	23.5	21.8	21.8	22.8	30.7	35.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	32.3	30.4	25.6	23.6	22.0	21.6	22.9	21.1	20.8	21.5	27.7	31.6			
	5% DB	34.4	31.5	25.3	21.7	20.4	20.2	20.6	18.9	19.3	20.7	28.8	33.8			
	MCWB	31.4	29.1	24.0	20.5	19.6	19.2	19.9	17.9	18.1	19.3	26.6	30.6			
	10% DB	33.3	29.9	23.4	19.4	17.9	17.5	16.8	16.0	17.0	18.6	27.6	32.6			
	MCWB	30.5	27.6	22.1	18.3	16.9	16.5	15.8	15.1	16.0	17.1	25.6	29.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	34.2	32.4	28.4	27.6	26.3	24.1	26.0	25.6	24.4	24.2	29.7	33.6			
	MCDB	36.2	34.1	29.2	27.8	27.0	24.5	26.3	26.0	25.1	24.7	31.3	36.2			
	2% WB	32.9	30.9	26.0	23.8	22.2	21.7	23.0	21.4	21.0	21.6	28.4	32.2			
	MCDB	34.8	32.6	26.5	24.2	22.7	22.3	23.6	21.7	21.7	22.5	29.7	34.3			
	5% WB	31.9	29.4	24.0	20.7	19.8	19.3	20.0	18.1	18.4	19.5	27.2	31.2			
(30) Mean Daily Temperature Range	MCDB	33.8	31.2	25.0	21.4	20.4	20.2	20.6	18.8	19.2	20.6	28.6	33.3			
	10% WB	31.0	27.8	22.1	18.4	17.0	16.6	16.0	15.1	15.9	17.1	25.9	30.2			
	MCDB	32.9	29.7	23.3	19.4	17.8	17.5	16.8	16.0	16.8	18.4	27.4	32.2			
	5% DB	6.9	8.9	9.0	8.5	8.3	8.3	8.7	9.0	8.8	9.3	8.6	7.4			
	MCDBR	7.3	8.1	8.5	8.2	7.5	7.6	8.1	8.2	7.0	7.7	8.6	7.7			
(35) Clear-Sky Solar Irradiance	MCWBR	6.2	7.7	8.6	8.4	7.7	7.6	8.1	8.3	7.0	7.5	7.6	6.6			
	5% WB	6.0	7.3	8.3	7.9	7.2	7.3	8.1	8.4	7.1	7.3	7.1	6.5			
	MCWBR	5.5	7.3	8.5	8.3	7.4	7.5	8.3	8.5	7.2	7.4	6.7	6.1			
	taub	0.264	0.249	0.227	0.185	0.126	N/A	0.130	0.184	0.215	0.226	0.249	0.264			
	taud	2.347	2.383	2.352	2.203	1.916	N/A	1.887	2.152	2.258	2.265	2.282	2.303			
(40) All-Sky Solar Radiation	Ebn at Noon	317	305	281	226	109	0	93	219	279	311	320	321			
	Edh at Noon	34	28	22	13	4	0	4	14	23	31	36	37			
(44) All-Sky Solar Radiation	RadAvg	2424	1699	903	328	51	0	19	182	666	1466	2303	2724			
	RadStd	100	87	61	22	3	0	1	9	39	57	112	141			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	-1.60	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page